

Sprog og oversættelse

Lexical analysis: From NFA to DFA; Minimizing a DFA

Part 4: About set equations

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Set equations

In this course, many of the algorithms that we meet, involve **set equations**. Some of the places where we see set equations are

- in the computation of ϵ -closures
- in top-down parsing, when we need to compute the FIRST and FOLLOW sets
- in data-flow analysis

There is a general approach for computing solutions to such equations. It works if the underlying universe for our sets is finite.

What is a set equation?

A set equation is an equation

$$X = F(X)$$

where X is an unknown, whose value is a set, and $F(X)$ is a set expression in which we can mention X .

The **universe** of our set equation is the set of elements that our solution can be a subset of.

An example

Assume that our universe is $U = \{a, b, c\}$. Here is a set equation over U .

$$X = \{a, b\} \cup X$$

Here $F(X) = \{a, b\} \cup X$.

A solution to this equation is $X = \{a, b\}$, since

$$\{a, b\} = \{a, b\} \cup \{a, b\}$$

In general there may be more than one solution to a set equation. Here another solution is $X' = \{a, b, c\}$, but we usually want to find the smallest solution.

An iterative algorithm

Suppose we consider sets over some finite universe U . In the case of ϵ -closures, U is the state set of our NFA.

Whenever we have a set equation of the form

$$X = F(X)$$

where $F(X)$ is a set expression in which X occurs, we can find a solution by finding the *sequence of approximations*

$$\emptyset, F(\emptyset), F(F(\emptyset)), \dots F^n(\emptyset)$$

This sequence will be increasing, if F satisfies certain conditions.

U is a finite universe, so at some point this sequence will converge: We find a k such that $F^k(\emptyset) = F(F^k(\emptyset))$. So the solution is

$$X = F^k(\emptyset)$$

Using the iterative algorithm

Remember the set equation over U where $U = \{a, b, c\}$.

$$X = \{a, b\} \cup X$$

and $F(X) = \{a, b\} \cup X$.

When we compute the sequence of iterations, we get

$$F(\emptyset) = \{a, b\} \cup \emptyset = \{a, b\}$$

$$F(F(\emptyset)) = \{a, b\} \cup (\{a, b\} \cup \emptyset) = \{a, b\}$$

so here we converge. And indeed this is our solution!

Computing the least solution

Not all set equations have a solution. If we use a set operator that is **non-monotonic**, we run into problems.

A set operator O is **monotonic** if whenever $X \subseteq Y$, then also $O(X) \subseteq O(Y)$.

Union and intersection are monotonic. If $X \subseteq Y$, then $X \cup A \subseteq Y \cup A$ and $X \cap A \subseteq Y \cap A$. But set complement and set difference are not.

The set operators that we use in our set equation must be monotonic; otherwise, the sequence of approximations will not be increasing.